

Twisted rational maps (with, R. Birkett, S. Zhao)

K field

- alg. closed
- $|\cdot|$ complete

$$\forall x \in K. |x| \geq 0 \quad |x| = 0 \iff x = 0$$

$$|xy| = |x| \cdot |y| \quad \forall x, y \in K$$

$$\underline{|x+y| \leq |x| + |y|}$$

K is non-archimedean if $\underline{|x+y| \leq \max\{|x|, |y|\}}$

K is archimedean if otherwise

Ostrowski's thm: $K = \underline{\mathbb{Q}}$ or non archimedean.

$\mathbb{Q}_p$ $\mathbb{C}\{\{t\}\}$ = completion of $\mathbb{C}\{\{t\}\}$ ^{formal} power series

$\mathbb{Q}_p$ $\mathbb{C}\{\{t\}\}$ $\mathbb{Z}$ $\mathbb{F}_p\{\{t\}\}$

- char $K = 0$

$f \in k(z)$ rational map $\deg f \geq 2$

$$f: \mathbb{P}_k^1 \rightarrow \mathbb{P}_k^1 \quad f^n = \underbrace{f \circ f \circ \dots \circ f}_n$$

Fatou set $F(f) = \{x \in \mathbb{P}_k^1 : \{f^n\} \text{ equicontinuous at } x\}$

Julia set $J(f) = \mathbb{P}_k^1 \setminus F(f)$.

Fact: • $k = \mathbb{C}$ $J(f) \neq \emptyset$. $F(f)$ may be empty

Pf: Suppose $J(f) = \emptyset$ $\{f^n\}$ equicontinuous in $\mathbb{P}_{\mathbb{C}}^1$

Arzela - Ascoli Thm: $\{f^n\}$ is a normal family

$$f^n \rightarrow f_{\infty} \leftarrow \text{rational} \\ \deg f_{\infty} = \infty \rightarrow \times$$

• $k = \text{non archimedean}$. k not locally compact, totally disconnected.

$J(f)$ may be empty $F(f) \neq \emptyset$.

proof of $\bar{F}(f) \neq \emptyset$

holomorphic index formula + nonrepelling periodic points
are in $\bar{F}(f)$

$$1 + \sum_{i=1}^{\deg f} \frac{1}{\lambda_i - 1} = 0$$

$$\Rightarrow \sum_{i=1}^{\deg f} \frac{1}{\lambda_i - 1} = -1$$

$$\Rightarrow \exists |\lambda_{i_0}| < 1$$

$\frac{dz}{f(z)-z}$ residue Thm
meromorphic.

$$|f^{n_0}(z_0)| \leq 1$$

Berkovich analytification.

$$\mathbb{A}^{1,an} = \left\{ \text{seminorms } \|\cdot\| \text{ on } K[z] \mid \|x\| = |x| \ \forall x \in K \subset K[z] \right\}$$

↑
Berkovich affine space

$$\text{Berkovich projective line } \mathbb{P}^{1,an} = \mathbb{A}^{1,an} \cup \{\infty\}$$

Berkovich: 4 types of points

$$p \in k[z]$$

type I $\cup$ type II always exists

type III $\cup$ type IV may not exist.

seminorm

$$\text{type I: } \{ \} \leftrightarrow |p(x)| \quad x \in k$$

norms

$$\text{type II: } \{ \} \leftrightarrow \sup_{x \in \overline{D}(x_0, r_0)} |p(x)|$$

$$\overline{D}(x_0, r_0) \subset k$$

$$x_0 \in k, \quad r_0 > 0 \\ r_0 \in |k^*|$$

$$\text{type III: } \{ \} \leftrightarrow \sup_{x \in \overline{D}(x_0, r_0)} |p(x)|$$

$$r_0 \notin |k^*|$$

$$\text{type IV: } \{ \} \leftrightarrow \lim_{n \rightarrow \infty} \sup_{x \in \overline{D}(x_n, r_n)} |p(x)|$$

$$\frac{\overline{D}(x_n, r_n) \supset \overline{D}(x_{n+1}, r_{n+1})}{r_n \in |k^*|}$$

$$\frac{1}{|A|^{1,an}} \rightarrow \mathbb{R}$$

$$\forall p \in k[z]$$

$$\{ \} \rightarrow \|p\|_g$$

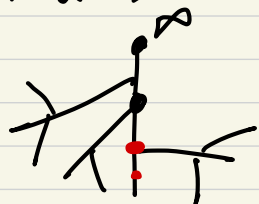
continuous.

weak topology

$$|p|^{1,an}$$

compact, connected path connected Hausdorff.

inclusion disks $\rightsquigarrow$ $|p|^{1,an}$ tree structure.

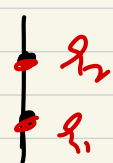


Rmk: $|p|^{1,an}$ is not metrizable.

$$\mathbb{P}^{1,an} \setminus \{\text{type I points}\} =: \mathbb{H} \leftarrow \text{Berkovich hyperbolic space.}$$

↑ hyperbolic metric

$$\begin{array}{ccc} f \searrow & & f \searrow \\ \mathbb{P}_K^1 & \hookrightarrow & \mathbb{P}^{1,an} \\ \text{type II} & & \text{dense} \\ & & \text{dense in } \mathbb{P}^{1,an} \end{array}$$



$$\rho(z_1, z_2) = \log \frac{\text{diam}(z_2)}{\text{diam}(z_1)}$$



$$\log \frac{r_2}{r_1}$$

$$\begin{array}{ccc} \mathbb{P}^{1,an} & \longrightarrow & \mathbb{P}^{1,an} \\ \mathfrak{z} & \longrightarrow & f(\mathfrak{z}) \end{array}$$

$$\|\phi\|_{f(\mathfrak{z})} := \|\phi \circ f\|_{\mathfrak{z}}$$

↑ rational / polynomial

$$f: \mathbb{P}^{1,an} \supset \{f^n\}$$

$$\text{Berkovich Julia set} : J_{\text{Ber}}(f) = \left\{ \mathfrak{z} \in \mathbb{P}^{1,an} \mid \forall U \ni \mathfrak{z} \bigcup_{n=0}^{\infty} f^n(U) \supset \mathbb{P}^{1,an} \setminus \{\text{finitely many points}\} \right\}$$

$$\text{Berkovich Fatou set} \quad F_{\text{Ber}}(f) = \mathbb{P}^{1,an} \setminus J_{\text{Ber}}(f)$$

Fact: $J_{\text{Ber}}(f) \cap \mathbb{P}_K^1 = J(f)$

$F_{\text{Ber}}(f) \cap \mathbb{P}_K^1 = F(f) \Rightarrow F_{\text{Ber}}(f) \neq \emptyset.$

Remark: "limit of ^{1 dimensional} archimedean system is nonarchimedean systems"
 Kiwi. deMarco, Favre Faber. Rivera-Letelier - - - .

twisted rational maps.

K field, $\text{Aut}(K) = \{ \text{continuous (field) automorphism of } K \}$

• $K = \mathbb{C}$ $\text{Aut}(K) = \{ \text{id}, z \rightarrow \bar{z} \}$

• $K = \text{nonarchimedean}$ $\text{Aut}(K)$ complicated

$K = \mathbb{L} = \overline{\mathbb{C}\{t\}}$

Deschamps: $\mathbb{C}\{t\} = \left\{ \sum_{n \geq n_0} a_n t^{\frac{n}{m}} \mid a_n \in \mathbb{C} \text{ m fixed} \right\}$

$$(1) \quad a_n \rightarrow \bar{a}_n$$

$$(2) \quad f^{\frac{1}{m}} \rightarrow f^{\frac{r}{m}} \quad r \in \mathbb{Q}.$$

$$(3) \quad f^{\frac{1}{m}} \rightarrow \sum_m f^{\frac{1}{m}} \quad \sum_m \text{ is a } m\text{-th root of unity}$$

admissible condition

$$(4) \quad f^{\frac{1}{m}} \rightarrow f^{\frac{1}{m}} \sum_{\alpha \in 1 + \mathcal{O}_K} (t^{\frac{1}{m}}) \quad f^{\frac{1}{km}} \rightarrow \sum_{km} f^{\frac{1}{km}}$$

$$\sum_{km}^k = \sum_m$$

upshot: $t \rightarrow at \quad a \in \mathbb{G} \setminus \{0\}$

$$\exists \sigma \in \text{Aut}(K) \text{ s.t. } \sigma(t) = at.$$

• $K = \mathbb{G}_p : \quad \underline{\text{Gal}(\bar{\mathbb{Q}}_p/\mathbb{Q}_p)} \subset \text{Aut}(\mathbb{G}_p)$

$$z \in K$$

$$f(z) \in K$$

$$\underline{z(f(z))} \in K$$

$$K = \text{non archimedean} \quad \text{Aut}(K) \supset \text{Aut}^*(K) := \{ z \in \text{Aut}(K) \}$$

$$\exists \lambda > 0 \text{ s.t.}$$

• $\lambda = 1$ valuation preserving cont. automorphism $|z(x)| = |x|^{\frac{1}{\lambda}}$

$$z \in \text{Aut}^*(K) \quad z: K \rightarrow K \quad z(\infty) = \infty$$

$$z: \mathbb{P}_K^1 \rightarrow \mathbb{P}_K^1$$

$$f \in K(z) \text{ rational map} \quad f \circ z(z)$$

• $K = \mathbb{C}$ $f \circ z = f$ or $f(\bar{z})$ ← anti-holomorphic.
 $(f \circ z)^2$ holomorphic.

• $K = \text{nonarchimedean}$ $f_z := f \circ z \rightsquigarrow \mathbb{P}_K^{1,q_n}$

$$z \curvearrowright K(z) \quad \hat{z} \cdot \phi(z) = \frac{z(a_n)z^n + z(a_{n+1})z^{n+1} + \dots + z(a_0)}{z(b_n)z^n + z(b_{n+1})z^{n+1} + \dots + z(b_0)} \in K(z)$$

$$\phi(z) = \frac{a_n z^n + a_{n+1} z^{n+1} + \dots + a_0}{b_n z^n + b_{n+1} z^{n+1} + \dots + b_0}$$

$$\begin{array}{ccc} \mathbb{P}_K^{1,q_n} & \longrightarrow & \mathbb{P}_K^{1,q_n} \\ \mathfrak{g} & \longrightarrow & f_z(\mathfrak{g}) \end{array}$$

$$\|\phi\|_{f_z(\mathfrak{g})} := \|\hat{z} \cdot \phi\|_{f(\mathfrak{g})} = \|(\hat{z} \cdot \phi) \circ f\|_{\mathfrak{g}}.$$

f_2 : twisted rational map.

Fact: f_2 restriction on $\mathbb{P}_K^1$ is z^1 of

twisted rational maps vs rational maps $K = \text{non-archimedean}$

"bad" { In general, f_2 is not rational

$(f_2)^n$ is not rational

f_2 is continuous.

$$f_2 \circ f_2 \circ f_2 \dots$$

$$f_2^2 = z^{-2} \circ f \\ = z^1 \circ z^{-1} \circ f$$

In general f_2 may have infinitely many fixed points (type I)

"good"

critical points / ramification points are same as f .

Rmk: "limit of 2-dimensional complex dynamics"

dynamics of f_2 Expect difference from dynamics of rational maps.

Find: dynamics similar as that for rational maps.
($\lambda \geq 1$) ($\lambda < 1$ very different)

2: $\lambda = 1$ (I.H. P) preserving
 $\lambda > 1$ expanding
 $\lambda < 1$ contracting.

Fatou set / Julia set

f_2^n

$J_{\text{Ber}}(f_2)$

$F_{\text{Ber}}(f_2)$

$J_I(f_2)$

$F_I(f_2)$

Thm: if $\lambda \geq 1$ $J_{\text{Ber}}(f_2) \cap \mathbb{P}_K^1 = J_I(f_2)$

$F_{\text{Ber}}(f_2) \cap \mathbb{P}_K^1 = F_I(f_2)$.

$$\lambda < 1 \quad F_I(f_2) \subseteq F_{\text{Ber}}(f_2) \cap P'_k$$

$$J_I(f_2) \supseteq J_{\text{Ber}}(f_2) \cap P'_k$$

$$F_I(f_2) \neq \emptyset ?$$

$$\lambda > 1 \quad F_I(f_2) \neq \emptyset$$

$$\lambda = 1 \quad F_I(f_2) \neq \emptyset$$

$$\lambda < 1 \quad F_I(f_2) = \emptyset \text{ may}$$

Thm: equidistribution:
 $d \cdot \lambda \geq 1$
 $d = \deg f$
 $\frac{1}{d^n} (f_2^n)^* \nu \rightarrow \mu_{f_2}$
 $\text{supp}(\mu_f) = J_{\text{Ber}}(f_2)$

f_2 same of f is same

$$f \in H(\mathbb{C})$$

fixed points

attracting
indifferent
repelling
saddle

Thm: If $\lambda \geq 1$. Let U be a fixed Fatou component (Berkovich)

Then ① attracting domain or

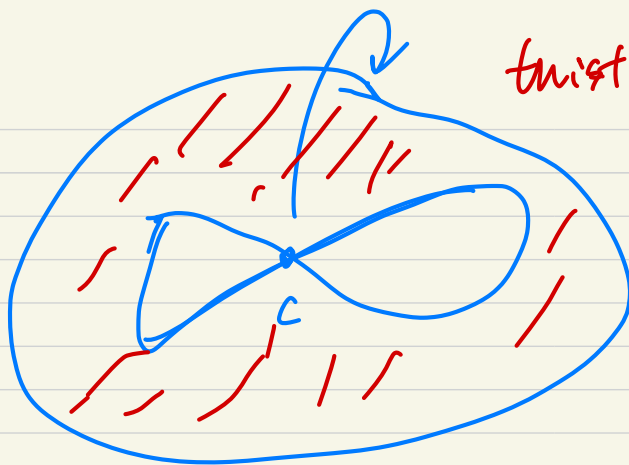
② Rivera domain with ∂U contains at most $(d-1)$ Julia periodic cycles

$$f \in K(z) \quad \text{same}$$

Thm: $f \in K(z) \quad K \subset \mathbb{C} \quad \text{discrete value group}$
 $\underline{z} \in \text{Aut}^*(K) \quad \exists \ell \supset \bar{K} \quad z(\ell) \subset \ell.$

All wandering Fatou components intersecting ℓ
 are attracted to some periodic cycle in Julia set J .

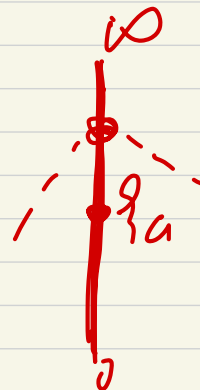
Thm: $f \in K[z] \quad f_2 \leftrightarrow \underline{\text{polynomials.}}$
 $\boxed{\lambda < 1}$



twist - deformation



TRUCO'S TREE



a

